



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Experiment No.: 3

1. Title: Implement control structures and form validation. Create a grading system based on user-entered marks.

2. Software/Tools Required: Visual Studio Code (VS Code) Google Chrome/web browser, HTML, JavaScript.

3. Theory:

A. Form Validation

Ensures user input is accurate and valid before processing.

- **Empty Check:** Ensures the input field is not blank.
- **Type Check:** Ensures input is a number using `isNaN()`.
- **Range Check:** Restricts marks within valid bounds.

B. Control Structures (`if...else if...else`)

Executes specific blocks of code depending on whether conditions evaluate to `true` or `false`. Used to map mark ranges to grades.

C. DOM Manipulation

- **Input:** `document.getElementById('marks').value` (Reads input).
- **Output:** `document.getElementById('result').innerText` (Displays output).



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4. Output:

Sonal's Student Grading System (24070521199):

Enter Marks:

Marks = 100
Grade = A

5. Result/Conclusion: By using control structures (`if...else if`) alongside form validation techniques, web applications can accurately process user inputs, prevent faulty data processing, and produce dynamic, real-time results such as student grading reports.